

CHETAN KULKARNI

Product & Digital Analytics | Open to Remote & California | 617-331-9347 | chetanmkulkarni@gmail.com |
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SUMMARY

Product data scientist with 8+ years delivering action-oriented insights that drive product strategy at scale. Expert in A/B test design and execution (20+ experiments), causal inference methods (staggered DiD, propensity score matching, pre-post analysis) for consumer product decision-making, and statistical modeling. Proven track record translating complex quantitative findings into measurable outcomes — from \$665M in new deposits to self-serve analytics tools that improve cross-functional data literacy. Proficient in SQL, Python, R, Looker, and Tableau. Comfortable with ambiguous, fast-paced environments.

PROFESSIONAL EXPERIENCE

Vanguard Group, Inc

Sep 2023 - Present

Lead Product Data Analyst / Data Scientist | Dallas, TX / San Diego, CA

- Developed action-oriented insights through observational causal analysis (staggered difference-in-differences, propensity score matching) to inform product strategy for Vanguard's digital consumer journeys, serving millions of members across web and mobile.
- Architected end-to-end experimentation framework; designed and executed 20+ A/B tests with power analysis and sample size estimation, driving a 5% reduction in support call rate (~\$250K+ annual savings) and 2–3% improvement in product engagement completion rate.
- Spearheaded Cash growth initiative: designed 5 targeted experiments in coordination with cross-functional partners (Product, Marketing, Finance), generating 19,000 new accounts in a single month — ~\$665M in new deposits and an estimated \$10M+ in annualized net interest revenue.
- Designed KPI frameworks and quarterly goal-setting metrics for digital product and marketing initiatives, delivering Looker and Tableau dashboards that enabled self-serve performance monitoring for senior stakeholders.
- Mentored junior analysts on statistical methodology, A/B test design, and causal inference; improved cross-functional data literacy by leading measurement education sessions with product and engineering teams.
- Served as analytics advisor to leadership, communicating statistical uncertainty and translating findings into resource allocation and strategic investment decisions.

Deloitte Consulting

Jul 2021 - Sep 2023

Senior Consultant - Analytics | San Diego, CA

- Applied causal inference and A/B testing — including non-inferiority experiments and staggered difference-in-differences — to measure impact of product design changes on user engagement and conversion for client financial web platforms.
- Built self-serve analytics tools and interactive dashboards in Tableau and Looker; improved cross-functional data literacy by enabling product and design teams to independently monitor key product metrics.
- Developed KPI frameworks and measurement strategies for digital products, aligning analytics output to quarterly OKRs and business goals.
- Built ETL pipelines and data infrastructure using Apache Airflow and AWS Athena, supporting analytics at scale for client product workflows.
- Led investigative analytics projects from problem definition through delivery, producing actionable insights that shaped client product roadmap decisions.

Genentech, Inc (via Pro-Unlimited)

Jan 2020 - Jul 2021

Informatics Data Analyst | South San Francisco, CA

- Developed a 5-layer CNN to classify image quality in clinical pathology slides and an LBP-based quality application — core invention in US Patent (US2022/0318979 A1, approved 2025).
- Built production ML deployment pipeline with multi-GPU environment and Apache Airflow automation; computed statistical quality metrics integrated with Tableau dashboard for clinical review.
- Collaborated with cross-functional ML and pathology teams to translate model outputs into clinical workflow decisions.

Deloitte Consulting

Senior Consultant | Bangalore, India

Feb 2015 - Apr 2018

- Analyzed large-scale public sector healthcare insurance data using SQL and Python; designed anomaly detection pipelines and delivered data-driven insights for policy optimization.
- Led business case definition, scope planning, and data quality testing across cross-functional teams for enterprise analytics engagements.

TECHNICAL SKILLS

Experimentation: A/B Testing, Experiment Design, Power Analysis, Sample Size Estimation, Non-Inferiority Testing, Hypothesis Testing, Guardrail Metrics

Causal Inference & Stats: Staggered DiD, Propensity Score Matching, Pre-Post Analysis, Regression, Statistical Modeling, Forecasting, Bayesian Methods, Time Series Analysis

Languages: SQL (Expert), Python (Expert), R

ML & Modeling: Machine Learning, Deep Learning, Clustering, Uplift Modeling, CNN

Visualization & BI: Looker, Tableau, Data Studio

Data Engineering: Apache Airflow, dbt, ETL Pipeline Design, AWS (SageMaker, Athena), GCP, BigQuery

Frameworks: KPI Frameworks, Metrics Development, Measurement Strategy, Analytics Architecture, OKR Development

Tools: Adobe Analytics, Adobe Target, Git, Jupyter

EDUCATION

Master of Science in Data Analytics

Northeastern University, Boston, MA

Bachelor of Engineering in Information Science and Engineering

Visvesvaraya Technological University (VTU), Belgaum, India

PATENT

Image Quality Analysis for Artifact Detection in Pathology Slide Images (Approved 2025)

US Patent Application No. US2022/0318979 A1 | Published March 2022 | Genentech, Inc

Developed a patented method for detecting imaging artifacts in digital pathology slides, applying automated image processing techniques to identify and flag quality issues in whole-slide images used for clinical diagnosis.